

Course 6014D

Semester VI

Theory

4 Credits

# Environmental Laws & Policies

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes

**Civil Engineering**

Contact hours

**60**

Teaching scheme

**3:1:0:0**

Credits

**4**

CIE / ESE

**Theory CIE 40 / Theory ESE 60**

Self-learning

**5.5 h/week**

## Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6014D as Environmental Laws & Policies in Semester VI for Civil Engineering. The official SITTTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL NLSIU’s course on Constitution of India and Environmental Governance and polytechnic environmental legislation curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

**Source treatment:** Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Legal interpretations, compliance assessments, EIA reports and environmental policy decisions must be based on authorised legal counsel, official court orders, certified compliance data, current legislation and competent professional review.

**Verified source note:** Verified sources support environmental governance, constitutional rights (Article 21, 48A, 51A(g)), EPA 1986, CPCB/SPCB, EIA Notification 2006, UNFCCC/Paris Agreement, Biodiversity Act, Forest/Wildlife laws, waste rules and NGT. The four-module sequence is a learning path, not an asserted official module split.

### COURSE DASHBOARD

## What You Are Expected to Learn

**60**

Contact hours

**4**

Credits

**Theory CIE 40**

CIE

**Theory ESE 60**

ESE

## Course introduction

Environmental Laws & Policies studies the legal and regulatory framework for environmental protection in India and internationally. It develops the ability to analyze constitutional provisions for the environment, understand major environmental acts (Water, Air, EPA), evaluate the Environmental Impact Assessment (EIA) process, and navigate the roles of regulatory bodies like Pollution Control Boards and the National Green Tribunal while respecting the technical and legal boundaries of environmental governance.

**Malayalam support:** ു handbook ുുുുുുുുുുു syllabus topic ുുുുു ുുുുുുുുു; ുുുുുുുു principle, diagram/procedure, application, common mistake ുുുുുു ുുുുുുുു ുുുുുുുു.

## Prerequisite foundation

Environmental studies, basic social sciences, elementary law, engineering ethics and global awareness.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

### Teaching and assessment scheme

| L | T | P | S | Self-learning/week | Contact hours | Credits | CIE           | ESE           |
|---|---|---|---|--------------------|---------------|---------|---------------|---------------|
| 3 | 1 | 0 | 0 | 5.5 h/week         | 60            | 4       | Theory CIE 40 | Theory ESE 60 |

### STUDY METHOD

## How to Use This Handbook

### 1 • Understand

Read terms, conditions and purpose; make a short definition in your own words.

### 2 • Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

### 3 • Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

### 4 • Revise

Use questions, quick cards and common-mistake notes before examination.

#### OFFICIAL STATEMENTS

## Objectives and Course Outcomes

### Official course objectives

1. Explain environmental governance fundamentals, constitutional provisions and the right to a clean environment.
2. Explain major Indian environmental legislations, including the Water Act, Air Act and Environment Protection Act.
3. Explain the legal framework for forests, wildlife, biodiversity and the Environmental Impact Assessment (EIA) process.
4. Explain waste management rules, international environmental conventions and the role of the judiciary and NGT.

### Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

**Malayalam support:** CO നോട്ടീസ് exam-ന് നോട്ടീസ് നോട്ടീസ് നോട്ടീസ് നോട്ടീസ് നോട്ടീസ്. Define, explain, apply നോട്ടീസ് action words നോട്ടീസ്.

### Official Course Outcomes

| CO  | Official outcome                                                                                          | Bloom level |
|-----|-----------------------------------------------------------------------------------------------------------|-------------|
| CO1 | Explain the constitutional basis and foundational norms of environmental governance in India.             | Understand  |
| CO2 | Explain the scope, application and enforcement of major pollution control and protection acts.            | Understand  |
| CO3 | Explain the legal framework for forest conservation, wildlife protection and the EIA process.             | Understand  |
| CO4 | Explain international climate agreements, waste management rules and the role of environmental tribunals. | Understand  |

### Official CO-PO articulation matrix

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7-PO11 |
|-----|-----|-----|-----|-----|-----|-----|----------|
| CO1 | 3   | 2   | 2   | 2   | 2   | -   | -        |
| CO2 | 3   | 2   | 2   | 2   | 2   | -   | -        |
| CO3 | 3   | 2   | 2   | 2   | 2   | -   | -        |
| CO4 | 3   | 2   | 2   | 2   | 2   | -   | -        |

### SYLLABUS COVERAGE

## Curriculum Coverage Map

## All official major topics mapped to handbook chapters

| Module | Title                                             | Official major topics                                                                                                                                                                                                          | Hours            | CO  | Handbook section |
|--------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----|------------------|
| 1      | Environmental Governance and Constitutional Basis | Foundational norms of environmental governance; common law roots; constitutional scheme of environment (Article 48A, 51A(g)); the fundamental right to a clean environment (Article 21); communitarian management.             | Approx. 14 hours | CO1 | Module 1         |
| 2      | Major Pollution Control and Protection Acts       | The Water (Prevention and Control of Pollution) Act, 1974; The Air (Prevention and Control of Pollution) Act, 1981; The Environment (Protection) Act, 1986; Pollution Control Boards (CPCB and SPCB) - constitution and roles. | Approx. 16 hours | CO2 | Module 2         |
| 3      | Forest, Wildlife and Impact Assessment Law        | Indian Forest Act, 1927; Forest (Conservation) Act, 1980; Wildlife (Protection) Act, 1972; Biological Diversity Act, 2002; Environmental Impact Assessment (EIA) Law and the 2006 Notification.                                | Approx. 16 hours | CO3 | Module 3         |
| 4      | Waste Management, Conventions and Judiciary       | Waste management rules (Solid, Hazardous, Bio-medical); International conventions (UNFCCC, Kyoto, Paris); role of the Higher Judiciary (Public Interest Litigation); National Green Tribunal (NGT) - constitution and role.    | Approx. 14 hours | CO4 | Module 4         |

## LEARNING ROADMAP

### How the Modules Connect

#### Environmental Governance and Constitutional Basis

Foundational norms of environmental governance; common law roots; constitutional scheme of environment (Article 48A, 51A(g)); the fundamental right to a clean environment (Article 21); communitarian management.

CO1 · Approx. 14 hours

#### Major Pollution Control and Protection Acts

The Water (Prevention and Control of Pollution) Act, 1974; The Air (Prevention and Control of Pollution) Act, 1981; The Environment (Protection) Act, 1986; Pollution Control Boards (CPCB and SPCB) - constitution and roles.

CO2 · Approx. 16 hours

## Forest, Wildlife and Impact Assessment Law

Indian Forest Act, 1927; Forest (Conservation) Act, 1980; Wildlife (Protection) Act, 1972; Biological Diversity Act, 2002; Environmental Impact Assessment (EIA) Law and the 2006 Notification.

CO3 · Approx. 16 hours

## Waste Management, Conventions and Judiciary

Waste management rules (Solid, Hazardous, Bio-medical); International conventions (UNFCCC, Kyoto, Paris); role of the Higher Judiciary (Public Interest Litigation); National Green Tribunal (NGT) - constitution and role.

CO4 · Approx. 14 hours

### MODULE 1

## Environmental Governance and Constitutional Basis

Foundational norms of environmental governance; common law roots; constitutional scheme of environment (Article 48A, 51A(g)); the fundamental right to a clean environment (Article 21); communitarian management.

Approx. 14 hours

CO1

Understand · Apply

### Learning goals

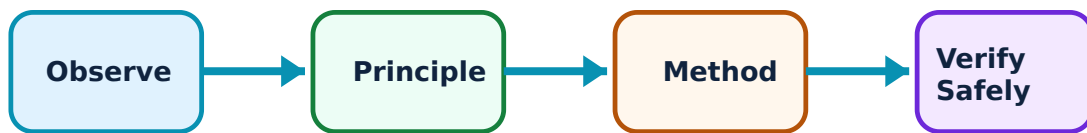
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Environmental Governance and Constitutional Basis: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Foundational norms and constitutional scheme–

Environmental governance in India is rooted in the Constitution. Article 48A directs the State to protect the environment, while Article 51A(g) makes it a fundamental duty of citizens. The Supreme Court has interpreted the Right to Life (Article 21) to include the right to a clean and healthy environment.

### Study and application method

Identify the two main constitutional articles that directly address environmental protection and explain the 'Right to Environment' under Article 21.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.



## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Identify the two main constitutional articles that directly address environmental protection and explain the 'Right to Environment' under Article 21.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept ഈ problem ഈ diagram / circuit / formula / procedure ഈ result ഈ application ഈ technical terms English-ഈ ഈ.

**Common mistake / precaution:** Constitutional interpretations are subject to judicial review. Do not treat a study summary as an authorised legal opinion on constitutional rights.

## 2. Environmental jurisprudence and governance–

Indian environmental law has evolved from common law roots to a comprehensive statutory framework. Governance involves a mix of centralized regulation and decentralized communitarian management, aimed at sustainable resource use and pollution prevention.

### Study and application method

Describe the transition from common law remedies to statutory environmental laws in India; explain the concept of 'communitarian management'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Describe the transition from common law remedies to statutory environmental laws in India; explain the concept of 'communitarian management'.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** concept diagram / circuit / formula / procedure application. Technical terms English-  
application. Technical terms English-  
application.

**Common mistake / precaution:** Governance structures vary by sector and region. Do not assume a uniform governance model applies to all natural resources without authorised policy analysis.

**Module 1 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

## MODULE 2

# Major Pollution Control and Protection Acts

The Water (Prevention and Control of Pollution) Act, 1974; The Air (Prevention and Control of Pollution) Act, 1981; The Environment (Protection) Act, 1986; Pollution Control Boards (CPCB and SPCB) - constitution and roles.

Approx. 16 hours

CO2

Understand · Apply

## Learning goals

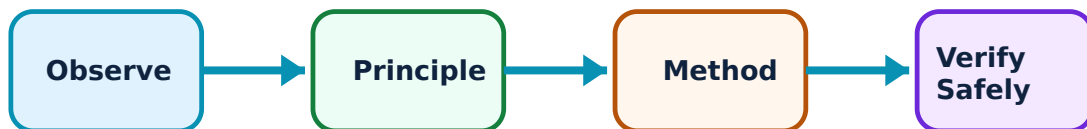
Identify core terms, explain the principle and complete the stated application or practical outcome.

## Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Major Pollution Control and Protection Acts: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Water and Air Acts–

The Water Act (1974) and Air Act (1981) were enacted to prevent and control pollution. They established the Central and State Pollution Control Boards (CPCB/SPCB) with powers to set standards, monitor compliance and take enforcement actions against violators.

### Study and application method

List four functions of the State Pollution Control Board (SPCB) under the Water Act; explain the purpose of 'consent to operate' for industries.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** List four functions of the State Pollution Control Board (SPCB) under the Water Act; explain the purpose of 'consent to operate' for industries.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept ഈ ചോദ്യത്തിന് ഉപയോഗിക്കണം. ഏതു diagram / circuit / formula / procedure ഉപയോഗിക്കണം. ഏതു result ഉണ്ടാകും application ഉപയോഗിച്ച് ഈ ചോദ്യത്തിന് ഉത്തരം എങ്ങനെ എഴുതണം. Technical terms English-ൽ എഴുതണം.

**Common mistake / precaution:** Pollution standards and compliance requirements are updated regularly. All industries must follow the current authorised standards issued by CPCB/SPCB.

## 2. The Environment (Protection) Act, 1986–

The EPA 1986 is an 'umbrella' legislation enacted after the Bhopal gas tragedy. It gives the Central Government wide powers to coordinate activities of various authorities, set environmental quality standards and regulate hazardous substances and industrial locations.

### Study and application method

Describe why the EPA 1986 is called an 'umbrella act'; identify two major powers of the Central Government under this act.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Describe why the EPA 1986 is called an 'umbrella act'; identify two major powers of the Central Government under this act.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept ഏതു device ഏതു arrangement. ഏതു diagram / circuit / formula / procedure ഏതു application. ഏതു result ഏതു application. Technical terms English-  
എന്നിവ ഉൾപ്പെടെ.

**Common mistake / precaution:** The EPA 1986 covers a wide range of environmental issues. Do not overlook specific notifications (e.g., on plastics or hazardous waste) issued under this act.

**Module 2 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

### MODULE 3

## Forest, Wildlife and Impact Assessment Law

Indian Forest Act, 1927; Forest (Conservation) Act, 1980; Wildlife (Protection) Act, 1972; Biological Diversity Act, 2002; Environmental Impact Assessment (EIA) Law and the 2006 Notification.

Approx. 16 hours

CO3

Understand · Apply

### Learning goals

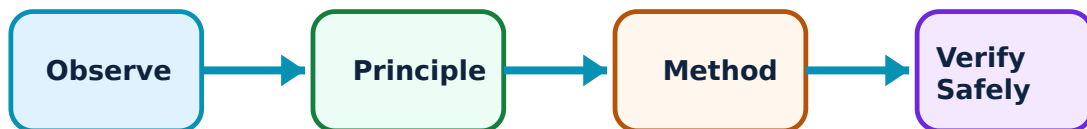
Identify core terms, explain the principle and complete the stated application or practical outcome.

## Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Forest, Wildlife and Impact Assessment Law: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Forest and wildlife legislation–

The Forest Acts regulate the use of forest land and prevent deforestation. The Wildlife Protection Act (1972) provides for the protection of wild animals, birds and plants, and the establishment of protected areas like National Parks and Sanctuaries.

### Study and application method

Compare the objectives of the Forest (Conservation) Act, 1980 and the Wildlife (Protection) Act, 1972; list two categories of protected areas.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Compare the objectives of the Forest (Conservation) Act, 1980 and the Wildlife (Protection) Act, 1972; list two categories of protected areas.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** കോൺസെപ്റ്റ് വ്യക്തമായി വിശദീകരിക്കുക. ഡയഗ്രാം / സിരക്യൂട്ട് / ഫോർമുല / പ്രോസീഡർ വ്യക്തമായി കാണിക്കുക. ഫലം വ്യക്തമായി പറയുക. ഫലം ഉപയോഗിച്ച് പ്രയോഗം കാണിക്കുക. Technical terms English-ൽ ഉപയോഗിക്കുക.

**Common mistake / precaution:** Forest and wildlife laws involve complex land rights and conservation priorities. Do not engage in activities in forest or protected areas without authorised permits.

## 2. Environmental Impact Assessment (EIA)–

EIA is a mandatory process for large projects to predict and mitigate environmental impacts. The EIA Notification 2006 outlines the stages: Screening, Scoping, Public Consultation and Appraisal, leading to Environmental Clearance (EC).

### Study and application method

Outline the four main stages of the EIA process as per the 2006 Notification; explain the importance of 'Public Consultation' in EIA.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Outline the four main stages of the EIA process as per the 2006 Notification; explain the importance of 'Public Consultation' in EIA.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept ഏതു ഏതു ഏതു. ഏതു diagram / circuit / formula / procedure ഏതു. ഏതു result ഏതു application ഏതു. Technical terms English-ഏതു.

**Common mistake / precaution:** EIA reports are technical and legal documents. They must be prepared by accredited consultants and reviewed by authorised appraisal committees.

**Module 3 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

## MODULE 4

# Waste Management, Conventions and Judiciary

Waste management rules (Solid, Hazardous, Bio-medical); International conventions (UNFCCC, Kyoto, Paris); role of the Higher Judiciary (Public Interest Litigation); National Green Tribunal (NGT) - constitution and role.

Approx. 14 hours

CO4

Understand · Apply

## Learning goals

Identify core terms, explain the principle and complete the stated application or practical outcome.

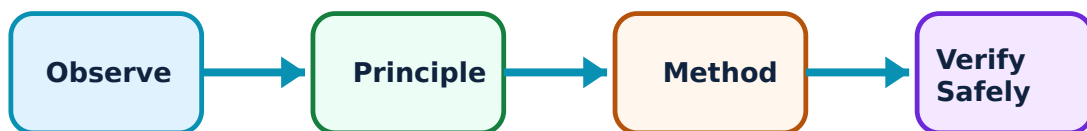


## Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Waste Management, Conventions and Judiciary: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Waste management and international law–

Specific rules under the EPA 1986 regulate the management of solid, hazardous and bio-medical wastes. Internationally, conventions like the Paris Agreement guide national climate actions and commitments to reduce greenhouse gas emissions.

### Study and application method

Identify the major responsibilities of municipal authorities under the Solid Waste Management Rules; explain India's commitment under the Paris Agreement.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Identify the major responsibilities of municipal authorities under the Solid Waste Management Rules; explain India's commitment under the Paris Agreement.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept ഈ ചോദ്യത്തിന് ഉപയോഗിക്കണം. ഈ diagram / circuit / formula / procedure ഉപയോഗിക്കണം. ഈ result ഉപയോഗിക്കണം application ഉപയോഗിക്കണം. Technical terms English-ൽ ഉപയോഗിക്കണം.

**Common mistake / precaution:** Waste management involves strict handling and disposal protocols. All waste generators must comply with current authorised rules to avoid legal penalties.

## 2. Judiciary and the National Green Tribunal–

The higher judiciary (Supreme Court/High Courts) plays a catalytic role through Public Interest Litigation (PIL). The National Green Tribunal (NGT) was established as a specialized body for the effective and expeditious disposal of environmental cases.

### Study and application method

Describe the role of the National Green Tribunal (NGT) in environmental adjudication; explain how 'Public Interest Litigation' (PIL) has helped environmental protection.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Describe the role of the National Green Tribunal (NGT) in environmental adjudication; explain how 'Public Interest Litigation' (PIL) has helped environmental protection.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept എങ്ങനെ എങ്ങനെ എങ്ങനെ. എന്ത diagram / circuit / formula / procedure എങ്ങനെ എങ്ങനെ. എങ്ങനെ result എങ്ങനെ application എങ്ങനെ എങ്ങനെ എങ്ങനെ. Technical terms English-  
എങ്ങനെ എങ്ങനെ.

**Common mistake / precaution:** Environmental litigation is a specialized legal field. All NGT petitions and judicial actions must be handled by authorised legal professionals.

**Module 4 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

## FORMULA AND PROCEDURE BANK

### Rapid Recall Bank

#### Recall 1

State symbols and SI units before substitution.

#### Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

### Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

### Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

### Recall 5

For a comparison: use construction, working, result, advantage and limitation.

### Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

#### DIAGRAM LIBRARY

### Diagram Library for Rapid Revision

#### Module 1: Environmental Governance and Constitutional Basis

Use the concept flow to revise observation, principle, method and verification.

#### Module 2: Major Pollution Control and Protection Acts

Use the concept flow to revise observation, principle, method and verification.

#### Module 3: Forest, Wildlife and Impact Assessment Law

Use the concept flow to revise observation, principle, method and verification.

## Module 4: Waste Management, Conventions and Judiciary

Use the concept flow to revise observation, principle, method and verification.

### SELF-LEARNING

## Official Self-Learning and Student Activities

### Structured activity

Map the foundational norms of environmental governance in a conceptual diagram.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Sketch the hierarchy of authorities responsible for pollution control in India.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Identify the stages of the EIA process for a hypothetical industrial project.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Compare two international climate agreements based on their targets and India's role.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

## Structured activity

Prepare a summary of a landmark environmental case decided by the Supreme Court of India.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

**Activity discipline:** The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

### EXAMINATION READINESS

## Assessment Methodology and Examination Pattern

| Official assessment structure |                            |                                                       |
|-------------------------------|----------------------------|-------------------------------------------------------|
| Component                     | Official mark / conversion | Student evidence                                      |
| Attendance                    | 5 marks                    | End-of-semester attendance component.                 |
| Self-learning activities      | 15 marks                   | Module-linked activities.                             |
| Series examinations           | 20 + 20 converted          | Two 50-mark series examinations.                      |
| CIE total                     | Theory CIE 40              | Continuous internal evaluation.                       |
| Written ESE                   | Theory ESE 60              | 2.5 hours; official Part A / Part B / Part C pattern. |

## Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

## Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

### EXAM PRACTICE

# Module-wise Questions with Answers

## Module 1 — Environmental Governance and Constitutional Basis

**Explain Foundational norms and constitutional scheme and state one correct method, application or precaution. –**

Environmental governance in India is rooted in the Constitution. Article 48A directs the State to protect the environment, while Article 51A(g) makes it a fundamental duty of citizens. The Supreme Court has interpreted the Right to Life (Article 21) to include the right to a clean and healthy environment.

**Answer framework:** Identify the two main constitutional articles that directly address environmental protection and explain the 'Right to Environment' under Article 21.

**Important point:** Constitutional interpretations are subject to judicial review. Do not treat a study summary as an authorised legal opinion on constitutional rights.

**Explain Environmental jurisprudence and governance and state one correct method, application or precaution. –**

Indian environmental law has evolved from common law roots to a comprehensive statutory framework. Governance involves a mix of centralized regulation and decentralized communitarian management, aimed at sustainable resource use and pollution prevention.

**Answer framework:** Describe the transition from common law remedies to statutory environmental laws in India; explain the concept of 'communitarian management'.

**Important point:** Governance structures vary by sector and region. Do not assume a uniform governance model applies to all natural resources without authorised policy analysis.

## Module 2 — Major Pollution Control and Protection Acts

**Explain Water and Air Acts and state one correct method, application or precaution. –**

The Water Act (1974) and Air Act (1981) were enacted to prevent and control pollution. They established the Central and State Pollution Control Boards (CPCB/SPCB) with powers to set standards, monitor compliance and take enforcement actions against violators.

**Answer framework:** List four functions of the State Pollution Control Board (SPCB) under the Water Act; explain the purpose of 'consent to operate' for industries.

**Important point:** Pollution standards and compliance requirements are updated regularly. All industries must follow the current authorised standards issued by CPCB/SPCB.

**Explain The Environment (Protection) Act, 1986 and state one correct method, application or precaution. –**

The EPA 1986 is an 'umbrella' legislation enacted after the Bhopal gas tragedy. It gives the Central Government wide powers to coordinate activities of various authorities, set environmental quality standards and regulate hazardous substances and industrial locations.

**Answer framework:** Describe why the EPA 1986 is called an 'umbrella act'; identify two major powers of the Central Government under this act.

**Important point:** The EPA 1986 covers a wide range of environmental issues. Do not overlook specific notifications (e.g., on plastics or hazardous waste) issued under this act.

### Module 3 — Forest, Wildlife and Impact Assessment Law

**Explain Forest and wildlife legislation and state one correct method, application or precaution. –**

The Forest Acts regulate the use of forest land and prevent deforestation. The Wildlife Protection Act (1972) provides for the protection of wild animals, birds and plants, and the establishment of protected areas like National Parks and Sanctuaries.

**Answer framework:** Compare the objectives of the Forest (Conservation) Act, 1980 and the Wildlife (Protection) Act, 1972; list two categories of protected areas.

**Important point:** Forest and wildlife laws involve complex land rights and conservation priorities. Do not engage in activities in forest or protected areas without authorised permits.

**Explain Environmental Impact Assessment (EIA) and state one correct method, application or precaution. –**

EIA is a mandatory process for large projects to predict and mitigate environmental impacts. The EIA Notification 2006 outlines the stages: Screening, Scoping, Public Consultation and Appraisal, leading to Environmental Clearance (EC).

**Answer framework:** Outline the four main stages of the EIA process as per the 2006 Notification; explain the importance of 'Public Consultation' in EIA.



**Important point:** EIA reports are technical and legal documents. They must be prepared by accredited consultants and reviewed by authorised appraisal committees.

## Module 4 — Waste Management, Conventions and Judiciary

**Explain Waste management and international law and state one correct method, application or precaution. –**

Specific rules under the EPA 1986 regulate the management of solid, hazardous and bio-medical wastes. Internationally, conventions like the Paris Agreement guide national climate actions and commitments to reduce greenhouse gas emissions.

**Answer framework:** Identify the major responsibilities of municipal authorities under the Solid Waste Management Rules; explain India's commitment under the Paris Agreement.

**Important point:** Waste management involves strict handling and disposal protocols. All waste generators must comply with current authorised rules to avoid legal penalties.

**Explain Judiciary and the National Green Tribunal and state one correct method, application or precaution. –**

The higher judiciary (Supreme Court/High Courts) plays a catalytic role through Public Interest Litigation (PIL). The National Green Tribunal (NGT) was established as a specialized body for the effective and expeditious disposal of environmental cases.

**Answer framework:** Describe the role of the National Green Tribunal (NGT) in environmental adjudication; explain how 'Public Interest Litigation' (PIL) has helped environmental protection.

**Important point:** Environmental litigation is a specialized legal field. All NGT petitions and judicial actions must be handled by authorised legal professionals.

### PRACTICE ONLY

## Practice Model Paper

**Important:** This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTTR model question paper.

## **Part A — short response**

1. State one key definition from Module 1: Environmental Governance and Constitutional Basis.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Major Pollution Control and Protection Acts.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Forest, Wildlife and Impact Assessment Law.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Waste Management, Conventions and Judiciary.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

## **Part B — structured explanation**

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

## **Part C — extended response / practical task**

1. Develop a complete answer or practical plan for: Foundational norms of environmental governance; common law roots; constitutional scheme of environment (Article 48A, 51A(g)); the fundamental right to a clean environment (Article 21); communitarian management..
2. Develop a complete answer or practical plan for: The Water (Prevention and Control of Pollution) Act, 1974; The Air (Prevention and Control of Pollution) Act, 1981; The Environment (Protection) Act, 1986; Pollution Control Boards (CPCB and SPCB) - constitution and roles..
3. Develop a complete answer or practical plan for: Indian Forest Act, 1927; Forest (Conservation) Act, 1980; Wildlife (Protection) Act, 1972; Biological Diversity Act, 2002; Environmental Impact Assessment (EIA) Law and the 2006 Notification..
4. Develop a complete answer or practical plan for: Waste management rules (Solid, Hazardous, Bio-medical); International conventions (UNFCCC, Kyoto, Paris); role of the Higher Judiciary (Public Interest Litigation); National Green Tribunal (NGT) - constitution and role..

## Quick Revision

### Module 1: Environmental Governance and Constitutional Basis

Foundational norms of environmental governance; common law roots; constitutional scheme of environment (Article 48A, 51A(g)); the fundamental right to a clean environment (Article 21); communitarian management.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 2: Major Pollution Control and Protection Acts

The Water (Prevention and Control of Pollution) Act, 1974; The Air (Prevention and Control of Pollution) Act, 1981; The Environment (Protection) Act, 1986; Pollution Control Boards (CPCB and SPCB) - constitution and roles.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 3: Forest, Wildlife and Impact Assessment Law

Indian Forest Act, 1927; Forest (Conservation) Act, 1980; Wildlife (Protection) Act, 1972; Biological Diversity Act, 2002; Environmental Impact Assessment (EIA) Law and the 2006 Notification.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 4: Waste Management, Conventions and Judiciary

Waste management rules (Solid, Hazardous, Bio-medical); International conventions (UNFCCC, Kyoto, Paris); role of the Higher Judiciary (Public Interest Litigation); National Green Tribunal (NGT) - constitution and role.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

**Common mistakes:** Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

**Exam checklist:** Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

## VOCABULARY

### Glossary

#### Important terms from the syllabus

| Term / topic                                 | One-line meaning                                                                                                  |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Foundational norms and constitutional scheme | Environmental governance in India is rooted in the Constitution.                                                  |
| Environmental jurisprudence and governance   | Indian environmental law has evolved from common law roots to a comprehensive statutory framework.                |
| Water and Air Acts                           | The Water Act (1974) and Air Act (1981) were enacted to prevent and control pollution.                            |
| The Environment (Protection) Act, 1986       | The EPA 1986 is an 'umbrella' legislation enacted after the Bhopal gas tragedy.                                   |
| Forest and wildlife legislation              | The Forest Acts regulate the use of forest land and prevent deforestation.                                        |
| Environmental Impact Assessment (EIA)        | EIA is a mandatory process for large projects to predict and mitigate environmental impacts.                      |
| Waste management and international law       | Specific rules under the EPA 1986 regulate the management of solid, hazardous and bio-medical wastes.             |
| Judiciary and the National Green Tribunal    | The higher judiciary (Supreme Court/High Courts) plays a catalytic role through Public Interest Litigation (PIL). |

## LEARNING RESOURCES

### References

#### Recommended environmental-law references

1. P. Leelakrishnan, Environmental Law in India.
2. Sairam Bhat, Natural Resources Conservation Law.
3. MoEFCC, Annual Reports and Policy Documents.

### **Approved public environmental-law sources**

- <https://nptel.ac.in/courses/129106002>
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These sources provide the verified study basis while official Course 6014D detail is unavailable; they do not replace official legal advice, official court orders, certified compliance reports or legal policy mandates.